



ELEVATE LAND SURVEYING

Lower Mainland Land Surveyors

FIELD SURVEY GUIDELINES

Our Company Goals: Professionalism, Integrity, Innovation

Common company phrases:

- 1.) Full Legal – Means getting all pins on the lot or getting pins that can define all boundaries(we will tie 2 monuments as well)
- 2.) Basic Legal-Means getting 3 or more pins(check with project manager if legal is appropriate if unsure)
- 3.) PM-Project manager
- 4.) TBM-Temporary Bench Mark
- 5.) EXV./LO-Excavation/Layout

General Field procedures

1. Refer to Job details or job PDF for instructions, check with project manager for questions
2. Cones should be used when near road ways
3. Field Notes should have title page that includes (job number, job address, job description, date, instrument type, crew names, weather)
4. We need original field notes, **NOT** re-written.
5. All notes must have PT#, Desc., Ht., Prism type, Comments(if any)
6. Notes should have record of setups, survey control, legal, and wrong ht., and any other critical comments
7. Point numbers for new jobs start at 100 (ex. Inst. @ 100 BS 101)
8. Use an elevation of 1000m or greater for first setup point created in a new job
9. Points 1-99 are saved for calc points for the office
10. 1000 pt. series are for GPS pts.(for desc. GPS Pt. 100 for shot on Pt. 100)
11. All legal tied in must have a legal sketch with point numbers(markup of legal plan is ok)
12. Detailed legal sketch showing NF needed for any full legal job
13. HI/HT should be recorded and inputted for all jobs
14. 99 Ht is used for reflectorless shots that do not have correct elevations
15. Independent checks are always needed when possible (ex. Chk 100 for check shot on pt. 100)
16. Checks to backsight at start and end of setups and or whenever needed if longer setup
17. If only one monument being tied, then take two shots with different rod heights(not needed for GPS)
18. All control set must start with 'O' (OCN, OSPK, ORBR, OCC etc.) see codelist
19. Tag numbers are noted in desc. (ex. OCN 2510) for a nail with tag #2510
20. Use a backsight tripod whenever possible(especially for longer setups and outset of work)
21. Use a round prism for longer shots and for backsights where possible/practical
22. Notes, data, and pictures to be uploaded by the end of day or job(whenever possible)

General Construction and or Layout procedures

1. Stakes for excavations marked with 'exv. 4' O/S'(typically we set 2' painted lathe)
2. For excavations Set benchmark at main floor or basement elevation
3. Tie the benchmark with **two** different heights as check(**critical**)

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4. Label TBM and take a picture (check with PM whether to label in Feet or metres)
5. For excavation Do **not** set stakes in fenced tree protection areas (Check with project manager)
6. For the Layout have a **recorded check** on the benchmark (**critical**)
7. Before the Layout check to see if excavated ground is about **0.45m** below basement elevation
8. For the Forms have a **recorded check** on the benchmark (if possible, critical if this is 2nd trip)
9. For the Forms shoot levels and locations at major corners
10. Measure any changes in level strips along walls
11. Measure curb wall strip if applicable
12. For the Forms get any posts
13. Measure up forms to confirm house dimensions
14. For gridlines jobs use a mini prism and or round prism where possible
15. Have a record of tasks done on title page for more complex construction jobs

General Topo procedures

1. Refer to job details for needed items(Always good to check with PM if in doubt)
2. Refer to our coding for proper description of features
3. Topo sketch is needed to outline critical features on site(building measurements, complex features etc.)
4. Manhole sketch of inverts (if needed)(use point numbers if needed to ensure no confusion)
5. Tree diameters must be given in centimeters (ie. tdec5 for a 5cm tree)
6. Round tree diameters to the nearest 5 (ie. 5, 10, 15, 20 etc.)
7. Width and direction of features noted in coding when tying only one side
8. Unique identification when tying different sides of features with same coding (ex. Bt1, bt2) for different sides of bank tops

General Equipment/Vehicle maintenance

1. Instrument cases must be **always** brought along with instrument.
2. Preference is that instruments are carried inside the case
3. Crew chief to use sound judgement on when to carry the instrument on a tripod (generally avoid on unstable, bushy, forested, trip hazard areas)
4. Instrument should not be carried by top handle (especially important for robotic instrument)
5. Do not put instrument and case in the back of the truck, should be kept in a stable/safe area
6. Items such as poles, prisms, controllers, tribrach's to be kept in appropriate protective cases
7. Calibration of the instrument needs to be done at least once a month
8. Controllers should be cleared out once a month(backup USB provided to a PM)
9. Prism poles need to be checked periodically(monthly) and adjusted if found out
10. Instruments need to be dried out and removed from the case after each day
11. **Any Damage** to critical equipment should be reported right away
12. Pin finders should also be dried out during extended wet periods
13. Crew chief responsible to check equipment and supplies and inform the manager if we are running short on supplies
14. Trucks needs to be cleaned and organized whenever possible (periodical car wash is encouraged)
15. Ensuring General vehicle maintenance is the responsibility of the crew chief(oil changes, repairs etc.)